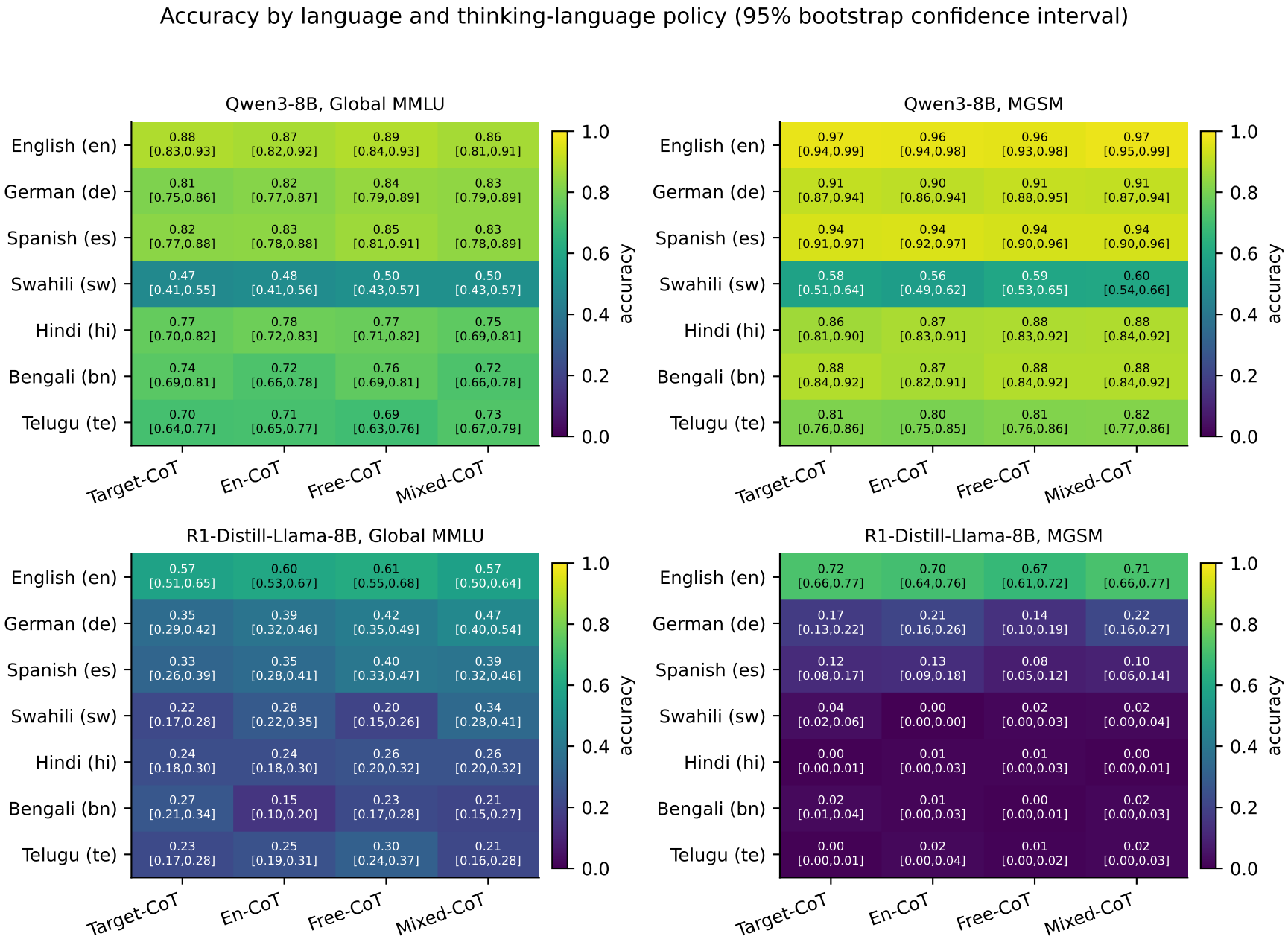
Accuracy by language and thinking-language policy (95% bootstrap confidence interval)
Qwen3-8B, Global MMLU
English (en) 0.88 [0.83,0.93] 0.87 [0.82,0.92] 0.89 [0.84,0.93] 0.86 [0.81,0.91]
German (de) 0.81 [0.75,0.86] 0.82 [0.77,0.87] 0.84 [0.79,0.89] 0.83 [0.79,0.89]
Spanish (es) 0.82 [0.77,0.88] 0.83 [0.78,0.88] 0.85 [0.81,0.91] 0.83 [0.78,0.89]
Swahili (sw) 0.47 [0.41,0.55] 0.48 [0.41,0.56] 0.50 [0.43,0.57] 0.50 [0.43,0.57]
Hindi (hi) 0.77 [0.70,0.82] 0.78 [0.72,0.83] 0.77 [0.71,0.82] 0.75 [0.69,0.81]
Bengali (bn) 0.74 [0.69,0.81] 0.72 [0.66,0.78] 0.76 [0.69,0.81] 0.72 [0.66,0.78]
Telugu (te) 0.70 [0.64,0.77] 0.71 [0.65,0.77] 0.69 [0.63,0.76] 0.73 [0.67,0.79]
Target-CoT
En-CoT
Free-CoT
Mixed-CoT
accuracy
1.0
0.8
0.6
0.4
0.2
0.0
Qwen3-8B, MGSM
English (en) 0.97 [0.94,0.99] 0.96 [0.94,0.98] 0.96 [0.93,0.98] 0.97 [0.95,0.99]
German (de) 0.91 [0.87,0.94] 0.90 [0.86,0.94] 0.91 [0.88,0.95] 0.91 [0.87,0.94]
Spanish (es) 0.94 [0.91,0.97] 0.94 [0.92,0.97] 0.94 [0.90,0.96] 0.94 [0.90,0.96]
Swahili (sw) 0.58 [0.51,0.64] 0.56 [0.49,0.62] 0.59 [0.53,0.65] 0.60 [0.54,0.66]
Hindi (hi) 0.86 [0.81,0.90] 0.87 [0.83,0.91] 0.88 [0.83,0.92] 0.88 [0.84,0.92]
Bengali (bn) 0.88 [0.84,0.92] 0.87 [0.82,0.91] 0.88 [0.84,0.92] 0.88 [0.84,0.92]
Telugu (te) 0.81 [0.76,0.86] 0.80 [0.75,0.85] 0.81 [0.76,0.86] 0.82 [0.77,0.86]
Target-CoT
En-CoT
Free-CoT
Mixed-CoT
accuracy
1.0
0.8
0.6
0.4
0.2
0.0
R1-Distill-Llama-8B, Global MMLU
English (en) 0.57 [0.51,0.65] 0.60 [0.53,0.67] 0.61 [0.55,0.68] 0.57 [0.50,0.64]
German (de) 0.35 [0.29,0.42] 0.39 [0.32,0.46] 0.42 [0.35,0.49] 0.47 [0.40,0.54]
Spanish (es) 0.33 [0.26,0.39] 0.35 [0.28,0.41] 0.40 [0.33,0.47] 0.39 [0.32,0.46]
Swahili (sw) 0.22 [0.17,0.28] 0.28 [0.22,0.35] 0.20 [0.15,0.26] 0.34 [0.28,0.41]
Hindi (hi) 0.24 [0.18,0.30] 0.24 [0.18,0.30] 0.26 [0.20,0.32] 0.26 [0.20,0.32]
Bengali (bn) 0.27 [0.21,0.34] 0.15 [0.10,0.20] 0.23 [0.17,0.28] 0.21 [0.15,0.27]
Telugu (te) 0.23 [0.17,0.28] 0.25 [0.19,0.31] 0.30 [0.24,0.37] 0.21 [0.16,0.28]
Target-CoT
En-CoT
Free-CoT
Mixed-CoT
accuracy
1.0
0.8
0.6
0.4
0.2
0.0
R1-Distill-Llama-8B, MGSM
English (en) 0.72 [0.66,0.77] 0.70 [0.64,0.76] 0.67 [0.61,0.72] 0.71 [0.66,0.77]
German (de) 0.17 [0.13,0.22] 0.21 [0.16,0.26] 0.14 [0.10,0.19] 0.22 [0.16,0.27]
Spanish (es) 0.12 [0.08,0.17] 0.13 [0.09,0.18] 0.08 [0.05,0.12] 0.10 [0.06,0.14]
Swahili (sw) 0.04 [0.02,0.06] 0.00 [0.00,0.00] 0.02 [0.00,0.03] 0.02 [0.00,0.04]
Hindi (hi) 0.00 [0.00,0.01] 0.01 [0.00,0.03] 0.01 [0.00,0.03] 0.00 [0.00,0.01]
Bengali (bn) 0.02 [0.01,0.04] 0.01 [0.00,0.03] 0.00 [0.00,0.01] 0.02 [0.00,0.03]
Telugu (te) 0.00 [0.00,0.01] 0.02 [0.00,0.04] 0.01 [0.00,0.02] 0.02 [0.00,0.03]
Target-CoT
En-CoT
Free-CoT
Mixed-CoT
accuracy
1.0
0.8
0.6
0.4
0.2
0.0